

To whom it may concern,

Please find enclosed our manuscript entitled “Strategic fisheries closure balances harvest opportunity and stock diversity in a large Alaska river basin” for consideration in the Journal of Applied Ecology. This study addresses an urgent management challenge for Pacific salmon in Alaska, where rapid declines in population health have heightened the need to evaluate whether existing management strategies can simultaneously support harvest opportunities and conserve the interpopulation biodiversity that underpins resilience.

Using otolith microchemistry, we reconstructed basin-wide patterns of spatial ecology in Chinook salmon entering the Kuskokwim River basin. In this system, alternative approaches such as genetic stock identification remain unavailable, and our analysis therefore demonstrates the value of otolith chemistry for inferring population-specific dynamics at fine spatial scales. We then evaluated how a currently implemented front-end fishery closure influences both the preservation of population diversity and the extent of foregone harvest opportunity.

Our results contribute to a growing understanding of the spatial ecology of Pacific salmon in large watersheds. Consistent with previous research, we found that early-arriving individuals migrate to the uppermost reaches of the basin. More broadly, our findings highlight the importance of analytical tools capable of reconstructing fine-scale ecological patterns to explicitly incorporate biodiversity considerations into management decisions. We argue that such tools are necessary to preserve the intra- and inter-population diversity that contributes to ecosystem resilience.

Given its applied focus, management relevance, and contribution to understanding how conservation actions shape ecological and evolutionary outcomes, we believe this manuscript is well aligned with the aims and scope of the Journal of Applied Ecology. This submission represents original work that has not been published elsewhere and is not under consideration by any other journal. We declare no conflicts of interest.

Thank you for considering our submission. We appreciate your time and look forward to the opportunity for review.

Sincerely,
Ben Makhoulf